Nathan Kingsley Chan

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EDUCATION

University of Illinois at Urbana-Champaign **Champaign, IL**
Grainger College of Engineering **Expected May 2028**
Bachelor of Science in Materials Science & Engineering | Hoeft Technology & Management Minor | James Scholar

RESEARCH INTERESTS

Design, processing, and characterization of high-performance metals and alloys for advanced manufacturing, energy, and industrial applications, with a focus on structure–property–processing relationships, failure analysis, and materials selection to improve durability, reliability, and performance.

Industry Experience

Dana Incorporated **Maumee, OH**
Materials Engineering Co-op **May 2026 - Aug 2026**

- Performed **materials and failure analysis** on 25+ production components for automotive OEMs (Ford, Toyota, Volvo, McLaren, Lamborghini), translating findings into technical reports for engineering teams
- Prepared **metallurgical specimens** (sectioning, mounting, grinding, polishing, acid etching) for stereo and optical microscopy per industry standards — direct experience in the sample-prep workflow used to characterize metal AM builds
- Conducted **hardness testing**, **chemical composition analysis (OES)**, and **magnetic particle inspection (MPI)** to verify material properties and detect/document surface defects against engineering specifications
- Analyzed **microstructures**, welds, and material properties against engineering drawings to support a zero-defect quality standard across a multi-OEM supplier network
- Investigated weld-quality effects of upstream process and supplier changes, **tying materials findings into process-improvement recommendations** that supported production throughput by **22%**
- Gained hands-on exposure to **SEM, EDS, and FTIR** characterization techniques;
- Implemented a lab-bench re-layout that 2x sample storage capacity, reducing retrieval time by **~50%**

Projects

Engineering Council, Engineering Open House Committee (EOH) **Champaign, IL**
Senior Director of Technology & Junior Director of Technology **Sept 2025 – Present**

- Leading full-stack development of eohillinois.org and the first interactive EOH app (React/Next.js, TypeScript, Firebase, Expo)
- Co-designed the backend and frontend of eohillinois.org, enabling the site to handle 30k+ annual visitors during EOH

Boring Illini **Champaign, IL**
Muck Removal Lead **May 2026- Present**

- Leading 8 members and cross-dimensional subsystem integrations across multiple projects for efficient excavation and soil-conditioning architecture design.

Project Lead ACP Dispensing System **Jan 2026 - May 2026**

- Led design of an active soil conditioning dispensing system to mitigate cohesive muck adhesion, cutterhead clogging, and torque spikes in medium-high cohesion soils
- Developed **high-level system architecture** and simplified flow path to reduce part count, cost, and failure modes
- Designed nozzle placement and mounting integration within vertical boring machine geometry; assisted **CAD development** of nozzle mounts for optimized injection and material breakup
- Responsible for generating BOM, engineering calculations, sponsor acquisition, and calibration/testing plan prior to procurement

Cutterhead Subteam Member **Sept 2025 – Dec 2025**

- Led experimental **torque and thrust analysis of cutterhead layouts and blade geometries**, measuring **revolutions required to climb through conditioned clay**; identified offset-control layouts that reduced required revolutions by **65%**
- Conducted a **blade material trade study** comparing **tool steels (S7, A2, D2, M2)** and **alloy steels (4140, 4130, 4340)** using hardness, toughness, machinability, and heat-treatment stability; selected **4140 steel** for optimal manufacturability, cost, and post-heat-treatment cutting performance. Assisted in selecting **thrust bearings** and housing to support increased Z-axis load, improving safety factor and reducing motor load.

RESEARCH EXPERIENCE

California Institute of Technology **Pasadena, CA**
Materials Science Researcher **Aug 2023 – May 2024**

- Designed and fabricated **micro-architected lithium-ion battery electrodes** using **nTopology** computational design and **hydrogel infusion additive manufacturing** (HIAM)
- Printed micro-architected organogels via **DLP vat photopolymerization**, followed by solvent exchange, aqueous metal salt infusion, calcination, and forming gas reduction
- Gained experience in **SEM** for morphology, and structural integrity analysis
- Contributed to fabrication of multi-material and alloy systems (Cu, CuNi, high-entropy alloys) through post-print material selection and thermal processing
- Applied principles of crystalline structure, defect mechanics, polymer behavior, and structure–property relationships to optimize lattice performance
- Collaborated with PhD mentor in weekly technical reviews; iterated lattice architectures based on experimental results and presented research findings

TECHNICAL SKILLS

- **Software Skills:** Fusion 360, Python, Java, JavaScript/TypeScript, MATLAB, nTopology, Excel, Data Modeling
- **Lab & Materials Characterization:** Metallography (sectioning, mounting, grinding, polishing, acid etching), optical/stereo microscopy, SEM, EDS, FTIR, OES, hardness testing, magnetic particle inspection (MPI), 3D printing (+ DLP/HIAM), cyclic voltammetry, soldering, manual milling, lathe/turning
- **Honors:** Eagle Scout, USA Water Polo Academic All-American, Barth Scholarship